

Mathematical Foundations of Political Methodology

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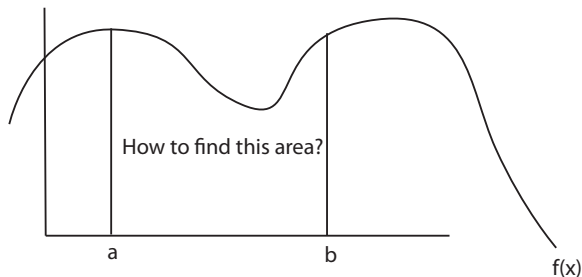
August 28th, 2017

- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

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Integral: Geometric Intuitions

“The opposite of differentiation,” “antiderivative,” or “taking an area”

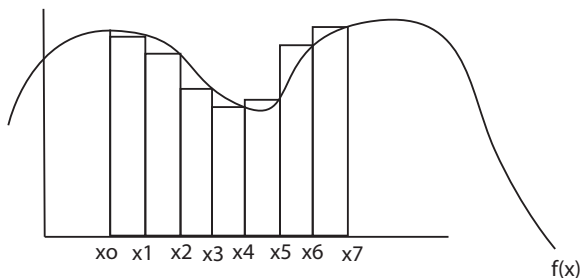


Integral: Geometric Intuitions

Let $a < b$ be real numbers and let f be a continuous function.

We want to find the area bounded by the curve $y = f(x)$, the vertical lines $x = a$ and $x = b$, and the x -axis. Let $a = x_0$ and $b = x_7$

How to find this area?

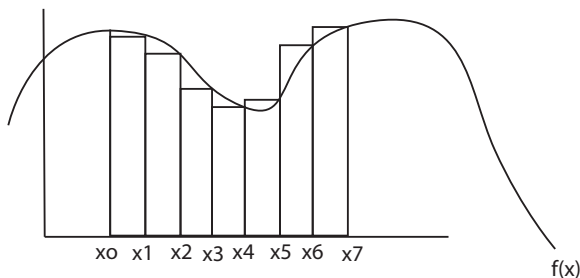


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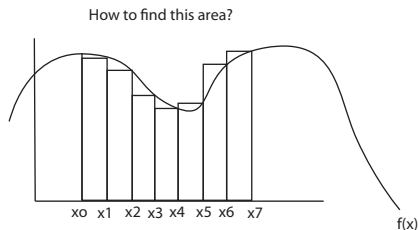
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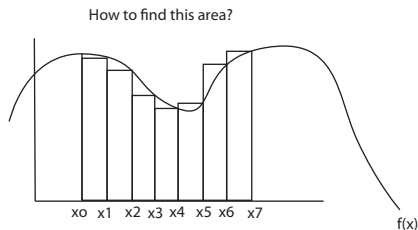


$$\sum_{i=1}^7 (x_i - x_{i-1}) f(x_i) \text{ or letting } \Delta x_i = x_i - x_{i-1} \text{ this approximation is}$$
$$\sum_{i=1}^7 f(x_i) \Delta x_i$$

Letting $\lim_{\Delta x_i \rightarrow 0}$ gives us the solution; it's written

$$\int_a^b f(x) dx$$

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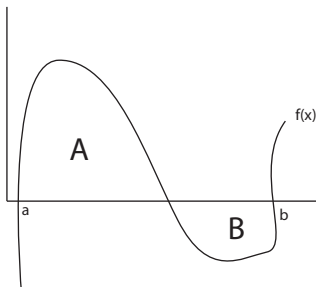
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Integral: Geometric Intuitions

Areas below the x -axis are negative:

$$\int_a^b f(x) = \text{Area}(A) - \text{Area}(B)$$



Integral: Definition

Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. We will define the Riemann Integral as $\int_a^b f(x)dx$. If this exists then we say f is *integrable* on $[a, b]$ and call $\int_a^b f(x)dx$ the *integral* of f .

Integral: Definition

Suppose $f : [0, 1] \rightarrow \frac{1}{x}$

$$\int_0^1 \frac{1}{x} dx$$

Then $\frac{1}{x}$ is not integrable on $[a, b]$ because the area that the integral would represent is infinite.

$$\begin{aligned} f(x) &= 1 \text{ if } x \text{ rational} \\ &= 0 \text{ if } x \text{ irrational} \end{aligned}$$

This function is called Dirichlet Function, it is not integrable, because every interval will contain a discontinuous jump

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Fundamental Theorem of Calculus

There is a connection between derivative and integral, the relationship is described by the Fundamental Theorem of Calculus. Before we introduce the Fundamental Theorem of Calculus, we introduce following theorem.

Theorem

Given a number a and a continuous function $f(x)$ we can define another function F such that

$$F(t) = \int_a^t f(x)dx \text{ for all } t$$

Then F is differentiable and $F'(t) = f(t)$ for all t

If you integrate a function and then differentiate the result, you retrieve the function that you started with $\Rightarrow$ “integration is the reverse of differentiation”

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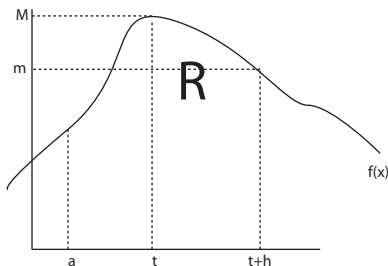
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Fundamental Theorem of Calculus



Proof: Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t . Then

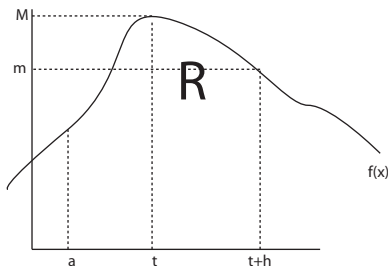
$$R = \int_a^{t+h} f(x)dx - \int_a^t f(x)dx = F(t+h) - F(t)$$

We also have that

$$hm \leq F(t+h) - F(t) \leq hM, \text{ or}$$

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

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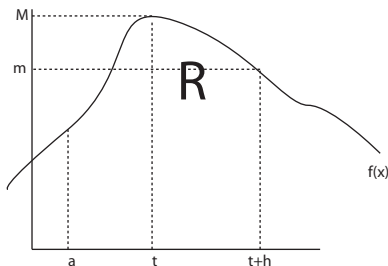
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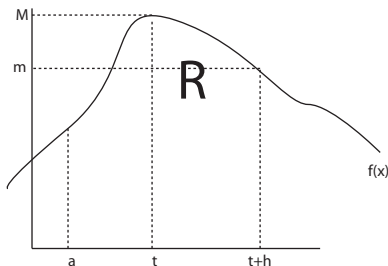
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As $h \rightarrow 0$ we have $m \rightarrow f(t) = M$. Thus,

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

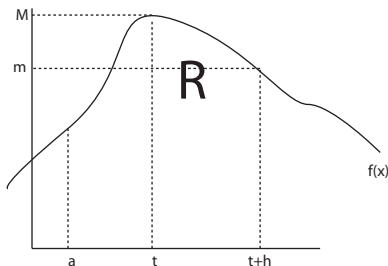
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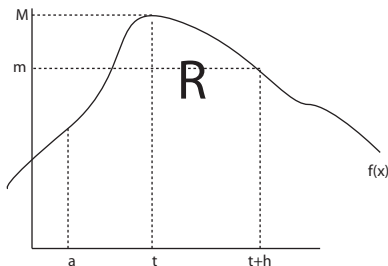
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$$H(t) = g(t) - \int_a^t g'(x)dx$$

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Therefore H is a constant, and $H(a) = H(b)$ for all a, b .

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Apply Fundamental Theorem of Calculus

$$\int_a^b f'(x) dx = f(b) - f(a)$$

- Calculate **antiderivative**
- Evaluate at b
- Evaluate at a

Some Classic Antiderivative Formulas

antiderivative = indefinite integral

$$\int 1 dx = x + c$$

$$\int k dx = kx + c$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\int \frac{1}{x} dx = \log x + c$$

$$\int e^x dx = e^x + c$$

$$\int a^x dx = \frac{a^x}{\log a} + c$$

Example 1: Uniform Distribution

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

What is the area under $f(x)$ from $[0, 1/2]$?

$$\begin{aligned} \int_0^{1/2} f(x) dx &= \int_0^{1/2} 1 dx \\ &= x \Big|_0^{1/2} \\ &= (1/2) - (0) \\ &= 1/2 \end{aligned}$$

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Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x) dx$

$$\begin{aligned}\int_2^t f(x) dx &= \int_2^t x dx \\&= \left. \frac{x^2}{2} \right|_2^t \\&= \frac{t^2}{2} - \frac{2^2}{2} \\&= \frac{t^2}{2} - \frac{4}{2} = \frac{t^2}{2} - 2\end{aligned}$$

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Integration Facts

Theorem

If $f_1, f_2 : [a, b] \rightarrow \mathbb{R}$ and f_1, f_2 are integrable on $[a, b]$, then

i) Consider the interval $[a, b]$ and $c \in [a, b]$. Then,

$$\begin{aligned}\int_c^c f_1'(x) dx &= f_1(c) - f_1(c) = 0 \\ \int_a^b f_1'(x) dx &= \int_a^c f_1'(x) dx + \int_c^b f_1'(x) dx \\ &= (f_1(c) - f_1(a)) + (f_1(b) - f_1(c)) \\ &= f_1(b) - f_1(a)\end{aligned}$$

Integration Facts

Theorem

If $f'_1, f'_2 : [a, b] \rightarrow \mathbb{R}$ and f'_1, f'_2 are integrable on $[a, b]$ and f'_1 has antiderivative f_1 and f'_2 has antiderivative f_2 , then

ii) For $c_1, c_2 \in \mathbb{R}$ then

$$\int_a^b (c_1 f'_1(x) + c_2 f'_2(x)) dx = c_1 \int_a^b f'_1(x) dx + c_2 \int_a^b f'_2(x) dx$$

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Indefinite Integrals

If $g(x)$ is a primitive of $f(x)$ (so that $g'(x) = f(x)$), then the set of all primitives of $f(x)$ is the set of all functions of the form $g(x) + C$, where C is any constant. C is known as the *constant of integration*

Let $f(x) = x + 7$; we know that a primitive of $f(x)$ is $g(x) = \frac{1}{2}x^2 + 7x$

It's also the case that $\tilde{g}(x) = \frac{1}{2}x^2 + 7x + 23$ is a primitive of f

Given a function f , the set of *all* its primitives is the *indefinite integral* of f , and is written

$$\int f(x)dx$$

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Indefinite Integrals: Examples

Example: The indefinite integral $\int 12x^2 dx = 4x^3 + C$, where C is an arbitrary constant

1

$$\int x^\alpha dx = \frac{1}{\alpha+1} x^{\alpha+1} + C \text{ if } \alpha \neq -1$$

2

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Indefinite Integrals: Examples

Example: The indefinite integral $\int 12x^2 dx = 4x^3 + C$, where C is an arbitrary constant

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Definite Integrals

With definite integrals we do not need to worry about the constant of integration: We know that

$$\int_a^b g'(x) dx = g(b) - g(a)$$

If we used $g(x) + C$ as our primitive instead we'd get

$$\int_a^b g'(x) dx = (g(b) + C) - (g(a) + C) = g(b) - g(a)$$

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